g2(0) cases

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(Dated: September 10, 2026)

abstract

We consider a negatively charged quantum dot (QD) with a resident electron embedded in a symmetric optical microcavity. The optically active transition couples the ground state, in which the QD hosts a single resident electron, to the negatively charged trion X^- with energy E_t, built of an electron singlet and a heavy hole. The energy of the fundamental cavity mode $\hbar\omega_0$ coincides with that of the trion transition, $E_t = \hbar\omega_0$, and the interaction with all other modes is neglected. The system is driven by a laser pulse of carrier frequency ω_L, detuned from the transition by $\Delta = \omega_L - \omega_0$, and is subject to a magnetic field B_x, perpendicular to the growth axis. We are interested in the non-resonant regime, in which the detuning exceeds both the spectral width Γ_L of the pump pulse and the width Γ of the cavity mode, $\omega_L - \omega_0 \gtrsim \Gamma_L, \Gamma$, as sketched in Fig. 1. As the pump pulse carries a large number of photons, this separation of scales allows to describe it as a classical field with the Rabi frequency $\Omega_R(t)$, whereas the cavity mode, which is not directly fed by the laser, contains only a few photons and therefore has to be retained at the operator level.

Within the rotating wave approximation the Hamiltonian of the system, which accounts for the coupling to the cavity mode, for the interaction with the driving laser field and for the interaction with the magnetic field, reads (hereafter $\hbar = 1$)

$$\begin{aligned}\hat{H} = E_t \sum_{\sigma=\uparrow,\downarrow} \hat{d}^\dagger_\sigma \hat{d}_\sigma + \omega_0 \sum_{\lambda=\pm} \hat{b}^\dagger_\lambda \hat{b}_\lambda \\ + g\left(\hat{d}^\dagger_\uparrow \hat{a}_\uparrow \hat{b}_+ + \hat{d}^\dagger_\downarrow \hat{a}_\downarrow \hat{b}_-\right) + g^*\left(\hat{d}_\uparrow \hat{a}^\dagger_\uparrow \hat{b}^\dagger_+ + \hat{d}_\downarrow \hat{a}^\dagger_\downarrow \hat{b}^\dagger_-\right) \\ + \Omega_R(t) \sum_{\sigma=\uparrow,\downarrow} \left(\mathrm{e}^{-i\omega_L t}\, \hat{d}^\dagger_\sigma \hat{a}_\sigma + \mathrm{e}^{i\omega_L t}\, \hat{d}_\sigma \hat{a}^\dagger_\sigma\right) \\ + \mu_B g_e B_x \left(\hat{a}^\dagger_\downarrow \hat{a}_\uparrow + \hat{a}^\dagger_\uparrow \hat{a}_\downarrow\right) + \mu_B g_t B_x \left(\hat{d}^\dagger_\downarrow \hat{d}_\uparrow + \hat{d}^\dagger_\uparrow \hat{d}_\downarrow\right)\end{aligned} \tag{1}$$

where the last two terms describe the Zeeman interaction in the transverse field. Here $\hat{a}^\dagger_\sigma$ is the creation operator of the resident electron, where $\sigma =\uparrow,\downarrow$ denotes the spin state $|\pm1/2\rangle$; $\hat{d}^\dagger_\sigma$ is the creation operator of the negatively charged trion, where σ denotes the heavy hole spin state $|\pm3/2\rangle$; $\hat{b}^\dagger_\pm = (\hat{b}^\dagger_H \pm i\hat{b}^\dagger_V)/\sqrt{2}$ is the creation operator of a cavity photon with circular polarization $\sigma_\pm$, expressed through the linearly polarized modes H and V; g is the effective QD–cavity coupling parameter; μ_B is the Bohr magneton, g_e and g_t are the in-plane g-factors of the resident electron and of the trion, respectively.

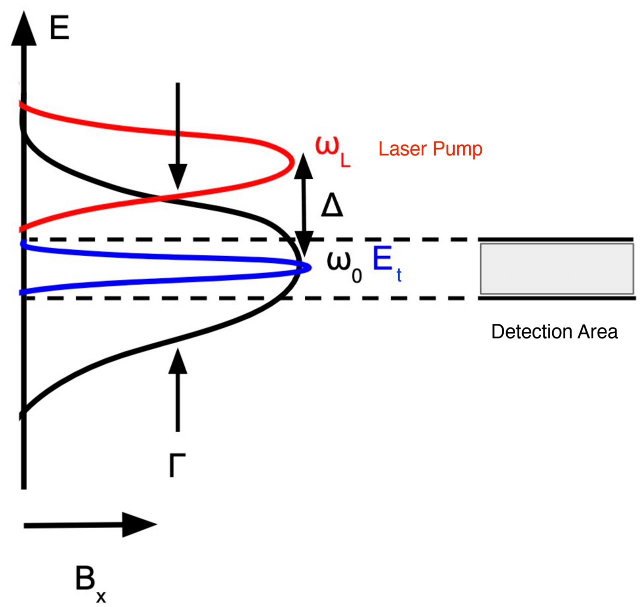

FIG. 1. Energy scheme of the system. The cavity mode of width Γ (black) is centered at ω_0 and is resonant with the trion transition E_t (blue). The pump pulse (red) has the carrier frequency ω_L and the spectral width Γ_L, and is detuned from the mode by Δ. The shaded box marks the spectral window of detection. The magnetic field B_x is applied perpendicular to the growth axis.

The optical selection rules fix the spin structure of the light–matter coupling: a σ_+ photon converts a resident electron $|+1/2\rangle$ into the trion with the heavy hole $|+3/2\rangle$, and a σ_- photon converts $|-1/2\rangle$ into $|-3/2\rangle$. Consequently, only the triads $(\hat{a}_\uparrow, \hat{d}_\uparrow, \hat{b}_+)$ and $(\hat{a}_\downarrow, \hat{d}_\downarrow, \hat{b}_-)$ are coupled, and the polarization of the emitted photon is rigidly tied to the spin state of the QD. The transverse field B_x, in turn, mixes the two spin branches, which is the mechanism that correlates the polarizations of the two emitted photons.

To describe the evolution of the system coupled to its environment, the Hamiltonian dynamics has to be supplemented by the dissipative terms of the Lindblad equation

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$$\partial_t \hat{\rho} = -i\left[\hat{H}, \hat{\rho}\right] + \sum_i \gamma_i \left(\hat{V}_i \hat{\rho} \hat{V}_i^\dagger - \frac{1}{2}\left\{\hat{V}_i^\dagger \hat{V}_i, \hat{\rho}\right\}\right) \equiv \hat{\mathcal{L}}\hat{\rho} \tag{2}$$

where $\{\cdot,\cdot\}$ is the anticommutator, and γ_i and $\hat{V}_i$ are the rate and the jump operator of the i-th channel. Three dissipation channels are retained.

The first one is the escape of photons from the cavity mode into the environment, which is the process that delivers the photons to the detector. Its jump operator and rate are

$$\hat{V}_{\text{phot}} = \hat{b}_\pm, \qquad \gamma_{\text{phot}} = \Gamma \tag{3}$$

where Γ is the width of the cavity mode.

The second channel is the pure dephasing of the optical transition induced by acoustic phonons, which is responsible for the damping of the Rabi oscillations. The corresponding jump operator is

$$\hat{V}_{\text{phon}} = \sum_{\sigma=\uparrow,\downarrow} \left(\hat{a}_\sigma^\dagger \hat{a}_\sigma - \hat{d}_\sigma^\dagger \hat{d}_\sigma\right), \qquad \gamma_{\text{phon}} = \alpha\, \Omega_R^2 \tag{4}$$

The third channel is the dephasing of the resident electron spin. It is described by the operator

$$\hat{V}_{\text{spin}\,i} = \begin{pmatrix}\hat{a}_\uparrow^\dagger & \hat{a}_\downarrow^\dagger\end{pmatrix} \hat{\sigma}_i \begin{pmatrix}\hat{a}_\uparrow \\ \hat{a}_\downarrow\end{pmatrix} + \begin{pmatrix}\hat{d}_\uparrow^\dagger & \hat{d}_\downarrow^\dagger\end{pmatrix} \hat{\sigma}_i \begin{pmatrix}\hat{d}_\uparrow \\ \hat{d}_\downarrow\end{pmatrix} \tag{5}$$

where $\hat{\sigma}_i$ with $i = x, y, z$ are the Pauli matrices acting in the spin space of the resident electron and of the trion. Since the magnetic field is directed along x, the operators $\hat{V}_{\text{spin}\,y}$ and $\hat{V}_{\text{spin}\,z}$ relax the transverse spin component, whereas $\hat{V}_{\text{spin}\,x}$ adds a pure dephasing of the longitudinal ones. The three rates are therefore expressed through the longitudinal and transverse spin relaxation times T_1 and T_2 as

$$\gamma_{\text{spin}\,x} = \frac{1}{T_1}, \qquad \gamma_{\text{spin}\,y} = \gamma_{\text{spin}\,z} = \frac{1}{T_2} \tag{6}$$

This channel destroys the coherence between the two spin branches and therefore directly degrades the polarization entanglement of the emitted photon pair.

Equation (2) is solved with the initial condition set by the state of the system before the arrival of the pump pulse, when the cavity mode is empty and the QD hosts only the resident electron. In the transverse magnetic field the eigenstates of the electron spin are the symmetric and antisymmetric superpositions $(|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$ split by the energy $2\mu_B g_e B_x$, and their populations follow the Boltzmann distribution at the lattice temperature T. The initial density matrix therefore reads

$$\begin{aligned}\hat{\rho}_0 = &\frac{f(B_x)}{2}\left(\hat{a}_\uparrow^\dagger + \hat{a}_\downarrow^\dagger\right)|0\rangle\langle 0|\left(\hat{a}_\uparrow + \hat{a}_\downarrow\right) \\ &+ \frac{f(-B_x)}{2}\left(\hat{a}_\uparrow^\dagger - \hat{a}_\downarrow^\dagger\right)|0\rangle\langle 0|\left(\hat{a}_\uparrow - \hat{a}_\downarrow\right)\end{aligned} \tag{7}$$

where $f(B_x) = \mathrm{e}^{\mu_B g_e B_x/T} \big/ \left(\mathrm{e}^{\mu_B g_e B_x/T} + \mathrm{e}^{-\mu_B g_e B_x/T}\right)$ and the temperature is measured in energy units.

The quantities measured in the experiment are the polarizations of the photons that reach the detector, so the observables have to be extracted from the projections of the full density matrix onto the sectors with a given number of emitted photons. Two routes bring a photon to the detector. It can leave the cavity already during the pulse, which corresponds to a quantum jump $\hat{b}_\lambda$ at a time $t_1 < \tau$, or it can be emitted after the end of the pulse by a trion that has survived until $t = \tau$ and then recombines. In the latter case the polarization of the photon is fixed by the trion spin through the selection rules, so that the spin index σ of an unrecombined trion can be read as the polarization index of the photon it is going to emit.

The evolution generated by Eq. (2) is conveniently written through the propagator $\mathrm{e}^{\hat{\mathcal{L}}t}$ of the Lindbladian. Along with the unconditional density matrix $\hat{\rho}(t)$, which starts from the thermal state (7), it is then natural to introduce the conditional density matrix $\hat{\rho}^{(\lambda\lambda')}(t|t_1)$, describing the system given that a photon has escaped from the mode at the instant t_1 and carrying the polarization indices of that photon,

$$\begin{aligned}\hat{\rho}(t) &= \mathrm{e}^{\hat{\mathcal{L}}t}\left(\hat{\rho}_0\right) \\ \hat{\rho}^{(\lambda\lambda')}(t|t_1) &= \mathrm{e}^{\hat{\mathcal{L}}(t-t_1)}\left(\hat{b}_\lambda\, \mathrm{e}^{\hat{\mathcal{L}}t_1}\left(\hat{\rho}_0\right)\hat{b}_{\lambda'}^\dagger\right)\end{aligned} \tag{8}$$

Here the operators $\hat{b}_\lambda$ and $\hat{b}_{\lambda'}^\dagger$ remove one photon from the mode and hand it over to the detector, after which the remaining part of the pulse drives the system further.

The one-photon density matrix is then a 2×2 matrix in the basis of the circular polarizations $\lambda = \pm$,

$$\begin{aligned}\left(\hat{\rho}_1\right)_{\lambda\lambda'} = &\langle 0|\,\hat{d}_\lambda \hat{\rho}(\tau)\hat{d}_{\lambda'}^\dagger\,|0\rangle \\ &+ \Gamma\int_0^\tau dt_1 \sum_{\sigma=\uparrow,\downarrow} \langle 0|\,\hat{a}_\sigma \hat{\rho}^{(\lambda\lambda')}(\tau|t_1)\hat{a}_\sigma^\dagger\,|0\rangle\end{aligned} \tag{9}$$

The first term collects the events in which the QD is left in the trion state at the end of the pulse and no photon has escaped before, while the second one collects the events in which a single photon has already been emitted during the pulse and the QD has returned to the ground state.

The two-photon sector is spanned by the states $\hat{b}_\lambda^\dagger \hat{d}_\sigma^\dagger\,|0\rangle$, in which one photon is already present while the trion is still excited and will emit the second one. Two distinct routes populate this sector. In the first one the photon is created by the QD–cavity coupling and stays in the mode until the end of the pulse without leaving the cavity, so that the whole dynamics remains coherent; this is the resonant scattering of the laser photons. In the second one the first photon has actually escaped from the cavity at t_1, and the remaining part of the pulse re-excites the QD, which then emits the second photon. Accordingly, the two-photon density matrix is a 4×4 matrix in the basis $|\lambda\sigma\rangle$ and consists of two contributions, $\hat\rho_2 = \hat\rho_2^{\text{scat}} + \hat\rho_2^{\text{re-ex}}$, with

$$\left(\hat\rho_2^{\text{scat}}\right)_{\lambda\sigma,\lambda'\sigma'} = \langle 0|\,\hat d_\sigma \hat b_\lambda \hat\rho(\tau)\hat b^\dagger_{\lambda'}\hat d^\dagger_{\sigma'}\,|0\rangle \tag{10}$$

$$(\hat\rho_2^{\text{re-ex}})_{\lambda\sigma,\lambda'\sigma'} = \Gamma\int_0^\tau dt_1\,\langle 0|\,\hat d_\sigma \hat\rho^{(\lambda\lambda')}(\tau|t_1)\hat d^\dagger_{\sigma'}\,|0\rangle$$

The probabilities of the one- and two-photon emission are the traces of the corresponding density matrices,

$$P_1 = \mathrm{Tr}\,\hat\rho_1, \qquad P_2 = \mathrm{Tr}\,\hat\rho_2 \tag{11}$$

The circular polarization of each emitted photon is produced by a single spin projection: the spin up state gives a σ_+ photon and the spin down state gives a σ_- one. The second photon is emitted when the trion recombines and the QD returns to its ground state, so that the resident electron is left in the spin state σ rigidly correlated with the polarization of that photon, and the pair stays entangled with the spin left behind in the QD. This correlation is removed by the subsequent spin relaxation. The spin relaxes towards the in-plane direction set by the transverse field, ending up in the thermal state built on the eigenstates $(|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$ with the populations $f(\pm B_x)$. The relaxation is slow on the scale of the whole emission process, and the coherence between the two spin branches that survives it is given by the same thermal factor that determines the initial spin coherence in Eq. (7). The two-photon polarization density matrix entering the entanglement measures is therefore

$$(\hat\rho_2)_{\lambda\sigma,\lambda'\sigma'} \to \frac{1}{P_2}\,w_{\sigma\sigma'}\,(\hat\rho_2)_{\lambda\sigma,\lambda'\sigma'} \tag{12}$$

$$w_{\sigma\sigma'} = \delta_{\sigma\sigma'} + (1-\delta_{\sigma\sigma'})\tanh\left(\frac{\mu_B g_e B_x}{T}\right)$$

normalized to unit trace. The populations, diagonal in σ, are not affected, while the coherences between the two spin branches are suppressed by $\tanh(\mu_B g_e B_x/T)$, so that the entanglement of the pair is limited by the polarization of the resident spin both before the pulse and after the emission.

In the polarization basis of the two photons, $|\lambda\lambda'\rangle$, the target state of the protocol is the maximally entangled Bell state

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|{+}{+}\rangle + |{-}{-}\rangle\right) \tag{13}$$

in which both photons carry the same circular polarization.

The quality of the generated state is characterized by two standard measures. The first one is the fidelity to the target Bell state,

$$F = \left\langle\Phi^+\right|\hat\rho_2\left|\Phi^+\right\rangle \tag{14}$$

which exceeds 1/2 only for entangled states and therefore already serves as an entanglement witness. The second one is the concurrence, which quantifies the entanglement itself,

$$C = \max\left(0,\,\lambda_1-\lambda_2-\lambda_3-\lambda_4\right) \tag{15}$$

where $\lambda_1 \ge \lambda_2 \ge \lambda_3 \ge \lambda_4$ are the eigenvalues of the matrix $\sqrt{\sqrt{\hat\rho_2}\hat{\tilde\rho}_2\sqrt{\hat\rho_2}}$, and $\hat{\tilde\rho}_2 = (\hat\sigma_y\otimes\hat\sigma_y)\,\hat\rho_2^*\,(\hat\sigma_y\otimes\hat\sigma_y)$ is the spin flipped density matrix, the complex conjugation being taken in the polarization basis. The concurrence vanishes for separable states and reaches unity for a maximally entangled one.

Before turning to the numerical solution let us look at the limiting form of $\hat\rho_2$. To first order in the small parameter $g\tau$, neglecting the phonon induced dephasing but keeping the spin relaxation of the resident electron after pulse pump, the two-photon density matrix generated by a 2π pulse reads, in the basis $|{+}{+}\rangle$, $|{+}{-}\rangle$, $|{-}{+}\rangle$, $|{-}{-}\rangle$,

$$\hat\rho_2^{(2\pi)} \approx (g\tau)^2\left(\frac{\Omega_R^2}{\Omega_R^2+\Delta^2}\right)^2\begin{pmatrix}1&0&0&\eta^2\\0&0&0&0\\0&0&0&0\\\eta^2&0&0&1\end{pmatrix} \tag{16}$$

where $\eta = \tanh\left(\mu_B g_e B_x/T\right)$. Only the $|{+}{+}\rangle \pm |{-}{-}\rangle$ states are populated, and the entire dependence on the magnetic field and on the temperature enters through the coherence η^2. The square appears because the spin coherence is picked up twice, once through the initial state (7) and once through the spin left behind after the emission (12). The factor $\Omega_R^2/(\Omega_R^2+\Delta^2)$ shows how the detuning influences the two-photon yield.

Equation (16) is a mixture of the two Bell states with thermal weights,

$$\hat\rho_2^{(2\pi)} \propto \frac{1+\eta^2}{2}\left|\Phi^+\right\rangle\left\langle\Phi^+\right| + \frac{1-\eta^2}{2}\left|\Phi^-\right\rangle\left\langle\Phi^-\right| \tag{17}$$

where $|\Phi^\pm\rangle = (|{+}{+}\rangle \pm |{-}{-}\rangle)/\sqrt{2}$. In the limit of a strong spin polarization, $\mu_B g_e B_x \gg T$, $\eta\to1$ and only $|\Phi^+\rangle$ survives, whereas for $\mu_B g_e B_x \ll T$ the two Bell states enter with equal weights and the pair is no longer entangled.

All the results presented below are obtained by a numerical solution of Eq. (2) with the following set of parameters: the cavity mode width $\Gamma = 50\,\mu$eV, the QD–cavity coupling $g = 25\,\mu$eV, the detuning $\Delta = 50\,\mu$eV,

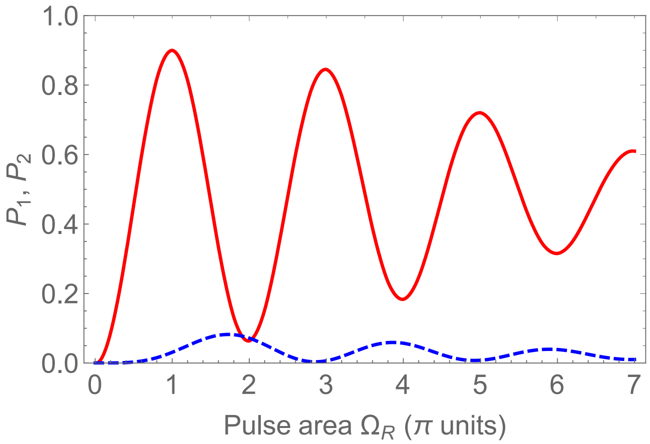

FIG. 2. Probabilities of the one-photon, P_1 (solid), and two-photon, P_2 (dashed), emission as functions of the pump pulse area.

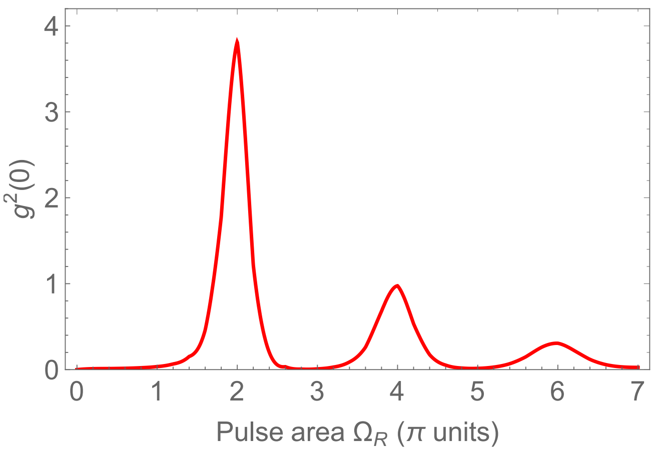

FIG. 3. Second-order correlation function at zero delay, $g^{(2)}(0)$, as a function of the pump pulse area.

the pulse duration $\tau = 13\,\mathrm{ps}$, the phonon coupling constant $\alpha = 0.15\,\mathrm{ps}$, the transverse magnetic field $B_x = 5\,\mathrm{T}$ and the lattice temperature $T = 1\,\mathrm{K}$. The pulse area $\theta = \int \Omega_R(t)\, dt$ is varied by changing the pump power at a fixed pulse duration, and is measured below in units of π.

The emission probabilities are shown in Fig. 2. The one-photon probability P_1 exhibits damped Rabi oscillations with maxima at odd and minima at even multiples of π, the damping being caused by the phonon-induced dephasing. The two-photon probability P_2 oscillates in antiphase and remains an order of magnitude smaller, reaching its maxima close to even pulse areas.

This antiphase behaviour is directly reflected in the second-order correlation function, Fig. 3. At odd pulse areas $g^{(2)}(0)$ nearly vanishes, so that the source emits single photons, whereas at even ones it displays pronounced bunching peaks, the largest reaching $g^{(2)}(0) \approx 3.8$ at $\theta = 2\pi$.

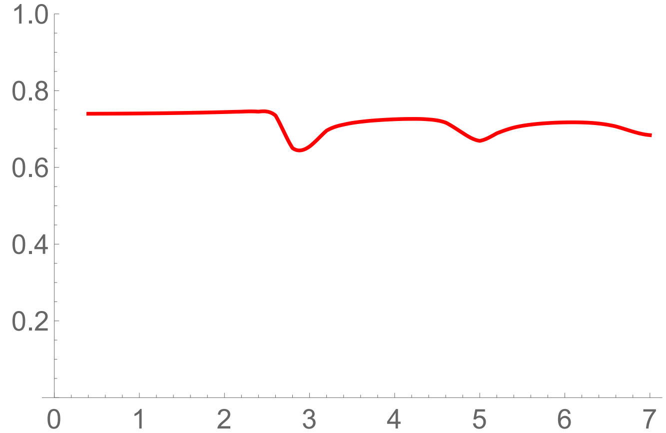

FIG. 4. Concurrence of the emitted two-photon polarization state as a function of the pump pulse area.

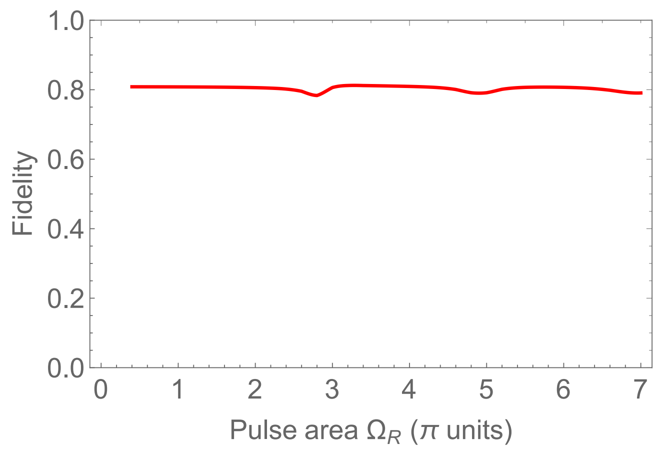

FIG. 5. Fidelity of the emitted two-photon polarization state to the Bell state $|\Phi^+\rangle$ as a function of the pump pulse area.

The polarization state of the emitted pair is characterized in Figs. 4 and 5. Both the concurrence and the fidelity to the Bell state $|\Phi^+\rangle$ are only weakly sensitive to the pump power, staying close to 0.7 and 0.8, respectively, over the whole range of pulse areas, with shallow minima near the odd ones. The fidelity exceeding $1/2$ confirms that the emitted pair is entangled for any excitation power.